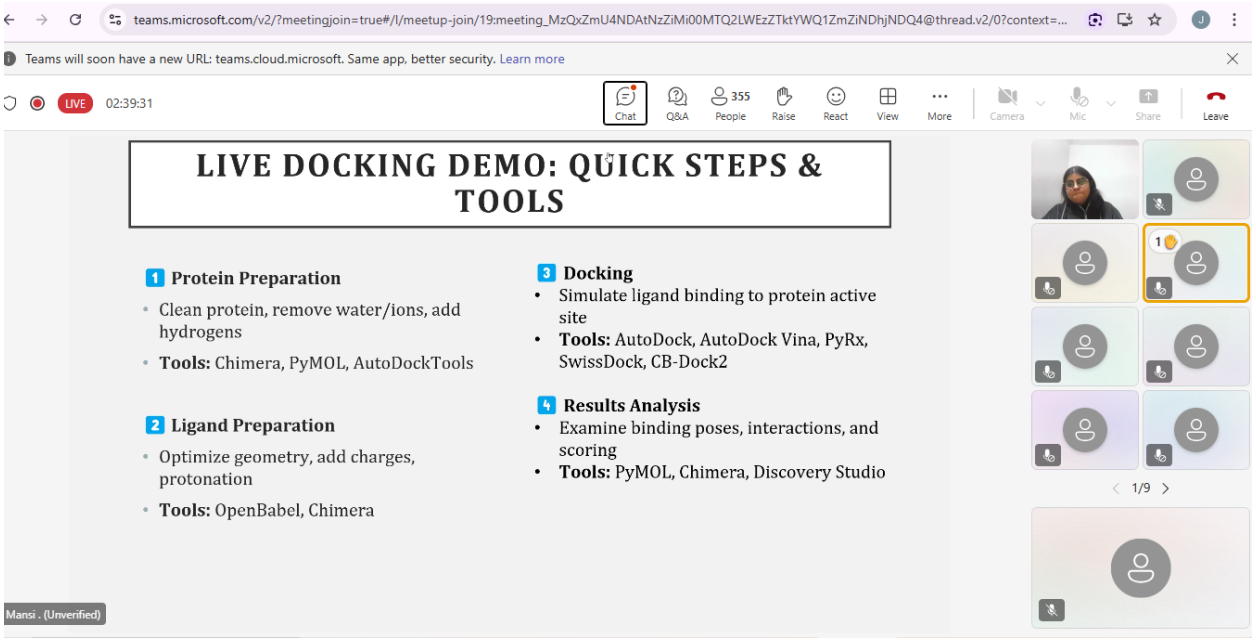
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355
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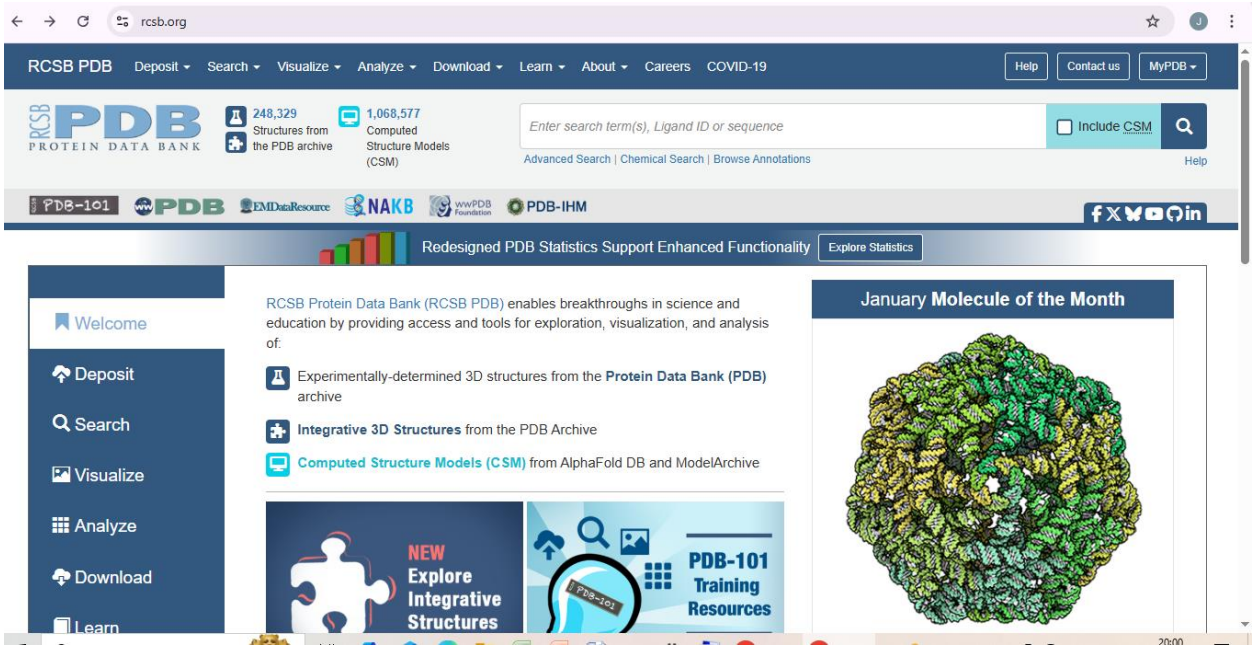
LIVE DOCKING DEMO: QUICK STEPS & TOOLS
1 Protein Preparation
Clean protein, remove water/ions, add hydrogens
Tools: Chimera, PyMOL, AutoDockTools
2 Ligand Preparation
Optimize geometry, add charges, protonation
Tools: OpenBabel, Chimera
3 Docking
Simulate ligand binding to protein active site
Tools: AutoDock, AutoDock Vina, PyRx, SwissDock, CB-Dock2
4 Results Analysis
Examine binding poses, interactions, and scoring
Tools: PyMOL, Chimera, Discovery Studio
1/9
Mansi . (Unverified)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rcsb.org/structure/1CA2

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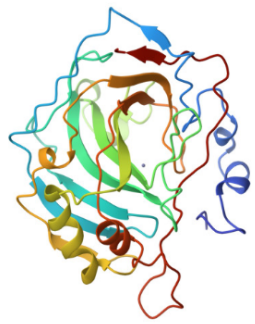
Biological Assembly 1



Explore in 3D: Structure | Sequence Annotations | Validation Report | Ligand Interaction (ZN)

Global Symmetry: Asymmetric - C1
Global Stoichiometry: Monomer - A1

Display Files | Download Files | Data API

1CA2 | pdb_00001ca2

REFINED STRUCTURE OF HUMAN CARBONIC ANHYDRASE II AT 2.0 ANGSTROMS RESOLUTION

PDB DOI: https://doi.org/10.2210/pdb1CA2/pdb Entry: 1CA2 **supersedes:** 1CAC

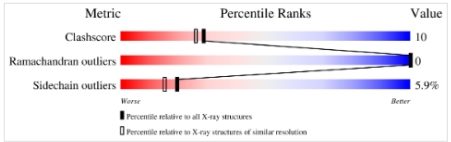
Classification: LYASE(OXO-ACID)
Organism(s): Homo sapiens
Mutation(s): No

Deposited: 1989-02-06 **Released:** 1990-01-15
Deposition Author(s): Eriksson, A.E., Jones, T.A., Liljas, A.

Experimental Data Snapshot

Method: X-RAY DIFFRACTION
Resolution: 2.00 Å

wwPDB Validation | 3D Report | Full Report



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Macromolecules

Find similar proteins by: Sequence (by identity cutoff) | 3D Structure

Entity ID: 1

Molecule	Chains	Sequence Length	Organism	Details	Image
CARBONIC ANHYDRASE II	A	259	Homo sapiens	**Mutation(s)**: 0 **EC:** 4.2.1.1 (PDB Primary Data), 4.2.1.69 (UniProt)	

UniProt & NIH Common Fund Data Resources

Find proteins for P00918 *(Homo sapiens)* Explore P00918 Go to UniProtKB: P00918

PHAROS: P00918 GTEx: ENSG00000104267

Entity Groups

Sequence Clusters	30% Identity, 50% Identity, 70% Identity, 90% Identity, 95% Identity, 100% Identity
UniProt Group	P00918

Sequence Annotations Expand

Reference Sequence 1CA2_1

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DISORDER

DISORDERED BINDING

PFAM

Small Molecules

Ligands 1 Unique

ID	Chains	Name / Formula / InChI Key	2D Diagram	3D Interactions
ZN Query on ZN Download Ideal Coordinates CCD File Download Instance Coordinates	B [auth A]	ZINC ION Zn PTFCDOFLOPIGGS-UHFFFAOYSA-N	Zn^{+2}	Interactions

Experimental Data & Validation

pubchem.ncbi.nlm.nih.gov/compound/1986

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COMPOUND

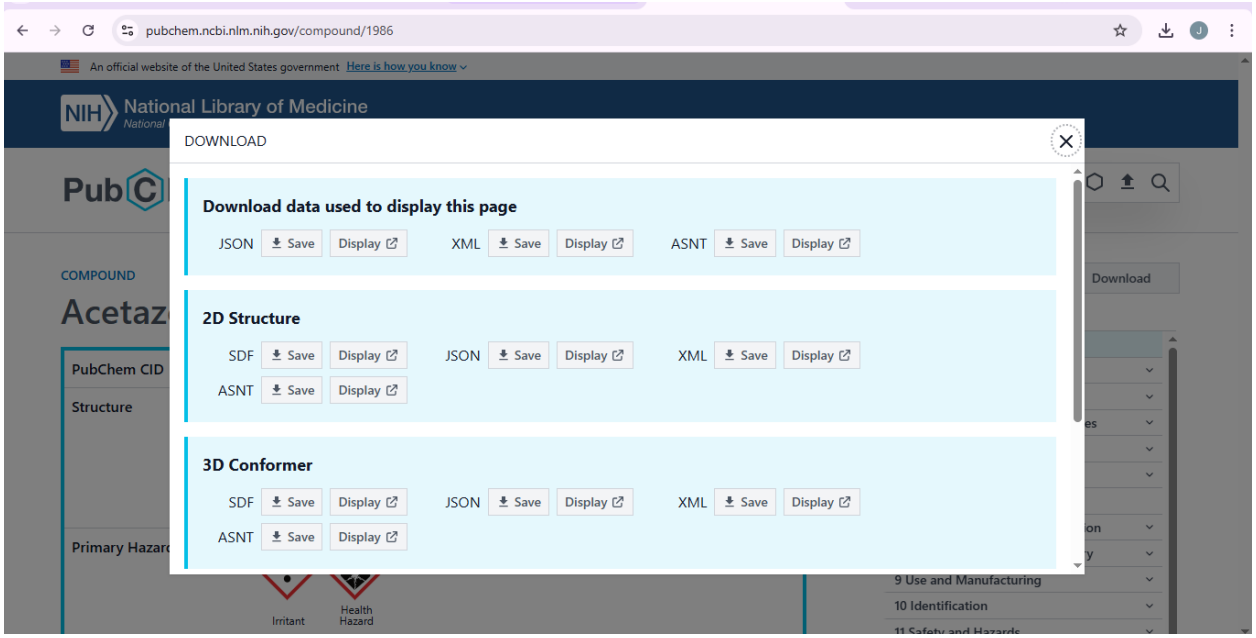
Acetazolamide

PubChem CID	1986
Structure	2D 3D Crystal
Primary Hazards	Irritant Health Hazard

Cite Download

CONTENTS

Title and Summary
1 Structures
2 Names and Identifiers
3 Chemical and Physical Properties
4 Spectral Information
5 Related Records
6 Chemical Vendors
7 Drug and Medication Information
8 Pharmacology and Biochemistry
9 Use and Manufacturing
10 Identification
11 Safety and Hazards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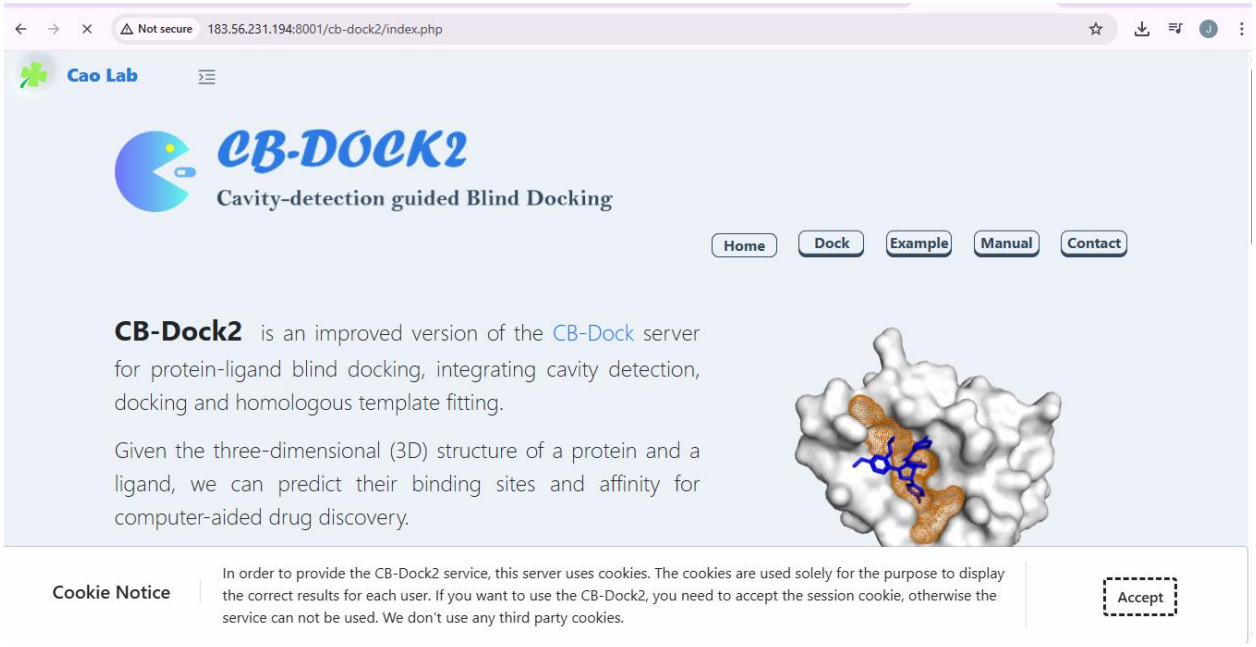
9 Use and Manufacturing
10 Identification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Cavity-detection guided Blind Docking
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Dock
Example
Manual
Contact
CB-Dock2 is an improved version of the CB-Dock server for protein-ligand blind docking, integrating cavity detection, docking and homologous template fitting.
Given the three-dimensional (3D) structure of a protein and a ligand, we can predict their binding sites and affinity for computer-aided drug discovery.
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Cao Lab

Home | Dock | Example | Manual | Contact

Upload Protein

Select a protein file (pdb format) [Example: MDM2.pdb (PDB: 4HG7)] [Submitted Protein: 9cd96a49-6b67-47a0-8052-549473a9013c.pdb]

```
COMPND    1CA2
AUTHOR    GENERATED BY OPEN BABEL 3.1.1
ATOM      1  N   HIS A   1      10.809  -0.520   7.814  1.00  0.00          N
ATOM      2  CA  HIS A   1       9.540   0.115   8.847  1.00  0.00          C
ATOM      3  C   HIS A   1       9.218  -0.939   9.627  1.00  0.00          C
ATOM      4  O   HIS A   1       9.362  -2.180   9.350  1.00  0.00          O
ATOM      5  CB  HIS A   1       9.034   1.208   8.184  1.00  0.00          C
ATOM      6  CG  HIS A   1       9.777   2.520   8.351  1.00  0.00          C
ATOM      7  ND1 HIS A   1       9.878   3.168   9.582  1.00  0.00          N
ATOM      8  CD2 HIS A   1      10.440   3.297   7.469  1.00  0.00          C
```

Protein uploaded!

Attention: The chains of the uploaded protein are broken at position A122(THR), A123(LYS)!

Remove file

Upload Ligand

Select a ligand file (mol2, mol, sdf, pdb format) [Example: Nutlin-3a.sdf] [Submitted Ligand: ligand.mol2]

Upload Local | Draw Ligand

```
@<TRIPOS>MOLECULE
CID 1986
19 19 1 0 0
PROTEIN
AMBER ff14SB


@<TRIPOS>ATOM
      1 S1        -3.0413    0.1096   -0.0849 S.o2      1 UNK   -0.0872
      2 S2        -0.3004    1.2561   -0.0551 S.2       1 UNK    0.1005
```

Ligand uploaded!

Remove file

Cao Lab

CB-DOCK2

Cavity-detection guided Blind Docking

Home | Dock | Results | Manual | Contact

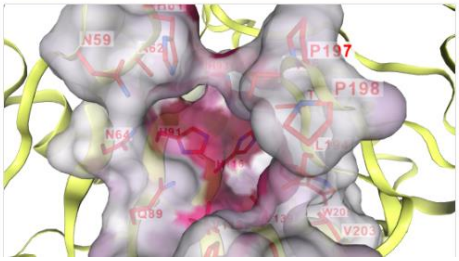
Structure-based cavity detection

Detected pockets from the input structure (Submitted Protein: 9cd96a49-6b67-47a0-8052-549473a9013c.pdb)

Tips: Click the table below to view the pocket information. Select the checkbox in the table below to mark the pockets you are interested in for blind docking. Click the residue in the sequence list to view the residue you are interested in.

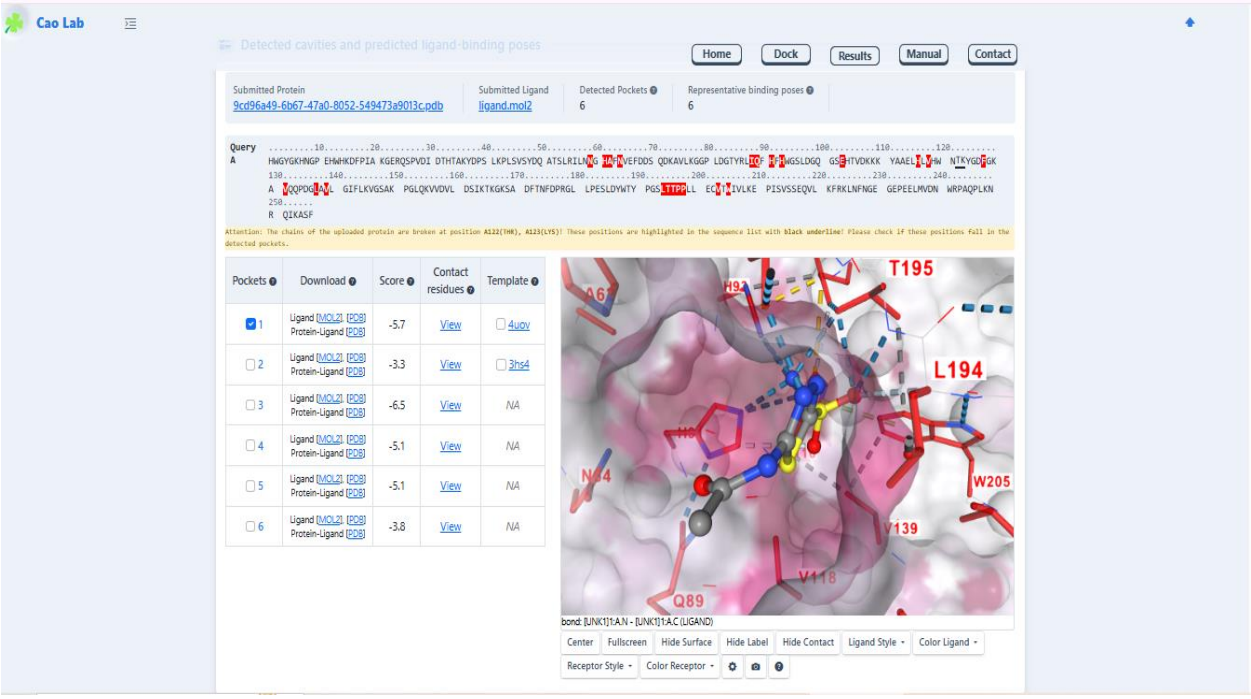
```
Query  .........10.........20.........30.........40.........50.........60.........70.........80.........90.........100........110........120........
A      HWGYGKHNGP EHWHKDFPIA KGERQSPVDI DTHTAKYDPS LKPLSVSYDQ ATSLRILNNG HAFNVEFDDS QDKAVLKGGP LDGTYRLIQF HFHWGSLDGQ GSEHTVDKKK YAAELHLVHW NTKYGDFGKA
       130........140........150........160........170........180........190........200........210........220........230........240........
       A  VGQPDEIAVL GIFLKVGSAK PGLQKVVDVL DSIKTKGKSA DFTNFDPRGL LPESLDYWTY PGSLTTPPLL ECVLWIVLKE PISVSSEQVL KFRKLNFNGE GEPEELMVDN WRPAQPLKN
       250.....
       R  QIKASF
```

Attention: The chains of the uploaded protein are broken at position A122(THR), A123(LYS)! These positions are highlighted in the sequence list with **black underline**! Please check if these positions fall in the detected pockets.



CurPocket ID	Cavity volume (Å^3)	Center (x, y, z)	Cavity size (x, y, z)
◉ C1	377	-5, 4, 15	7, 12, 10
○ C2	271	-12, 9, 29	10, 17, 13
○ C3	216	4, -5, 7	5, 13, 8
○ C4	148	-9, 9, 7	7, 8, 4
○ C5	136	-11, -11, -3	9, 6, 4

Download CurPockets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Home
Dock
Results
Manual
Contact
Submitted Protein
9cd96a49-6b67-47a0-8052-549473a9013c.pdb
Submitted Ligand
ligand.mol2
Detected Pockets
6
Representative binding poses
6
Pockets
Download
Score
Contact residues
Template
1 -5.7 View 4uov
2 -3.3 View 3hs4
3 -6.5 View NA
4 -5.1 View NA
5 -5.1 View NA
6 -3.8 View NA
T195
L194
W205
V139
V118
Q89
Center
Fullscreen
Hide Surface
Hide Label
Hide Contact
Ligand Style
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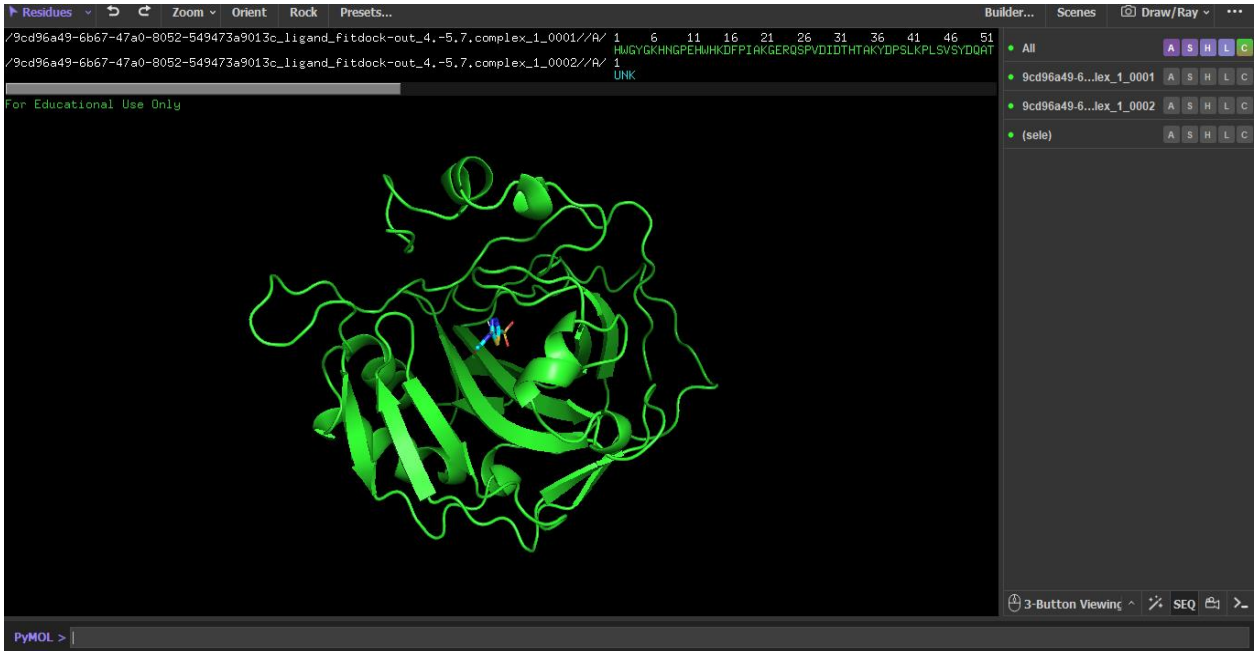
Receptor Style
Color Receptor


Residues
Zoom
Orient
Rock
Presets...
Builder...
Scenes
Draw/Ray
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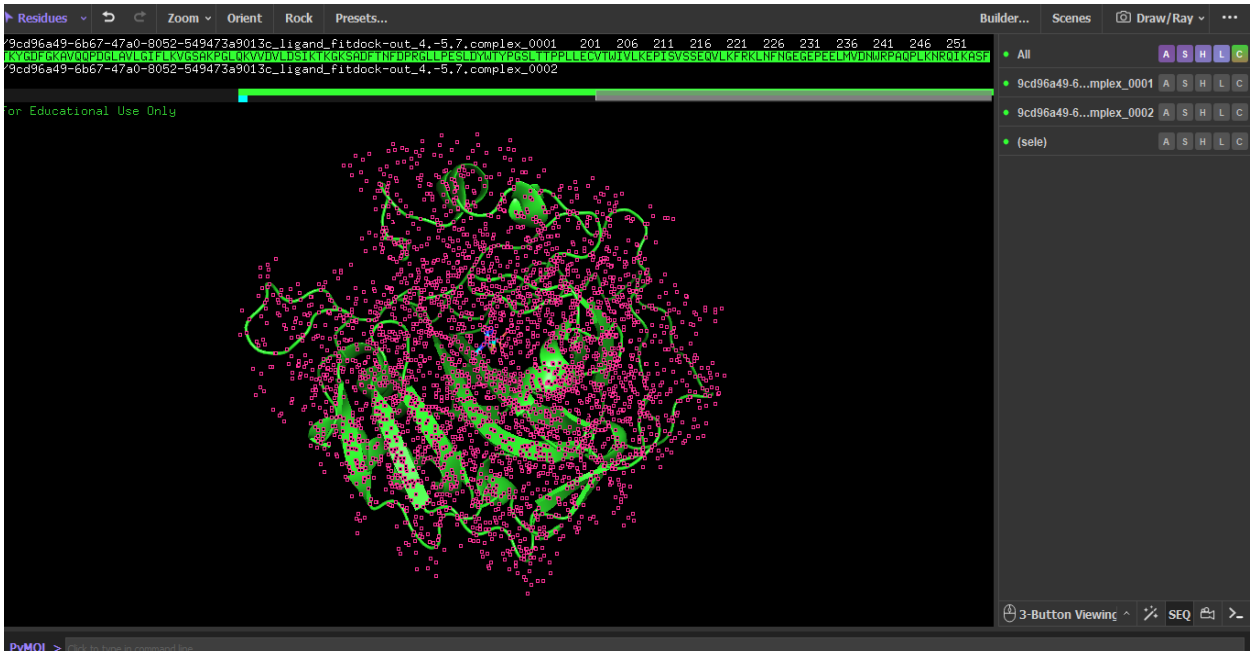
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Zoom
Orient
Rock
Presets...
Builder...
Scenes
Draw/Ray
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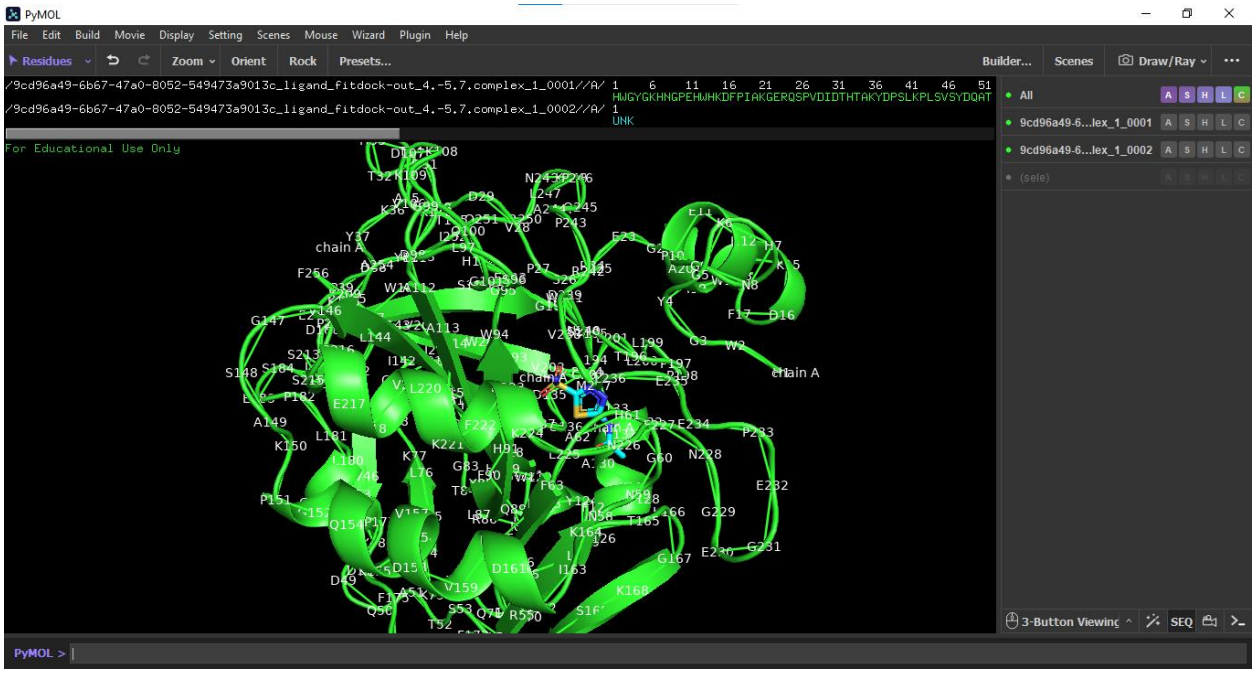
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File
Edit
Build
Movie
Display
Setting
Scenes
Mouse
Wizard
Plugin
Help
Residues
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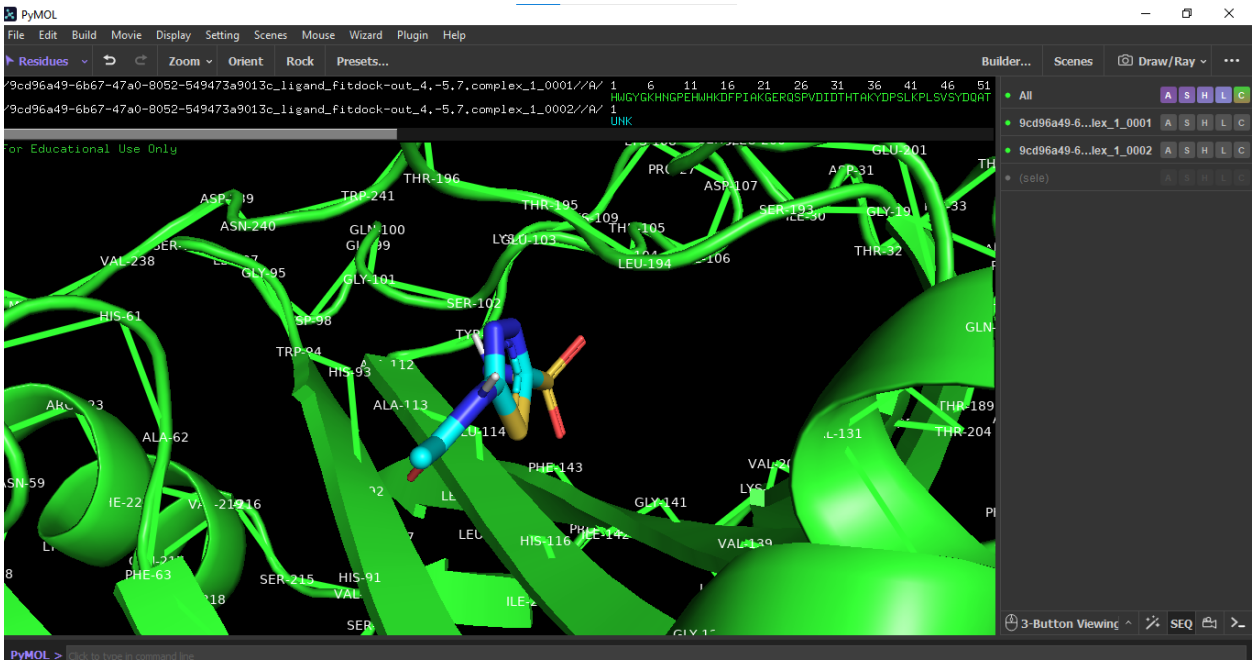
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File Edit Build Movie Display Setting Scenes Mouse Wizard Plugin Help
Residues
Zoom
Orient
Rock
Presets...
Builder...
Scenes
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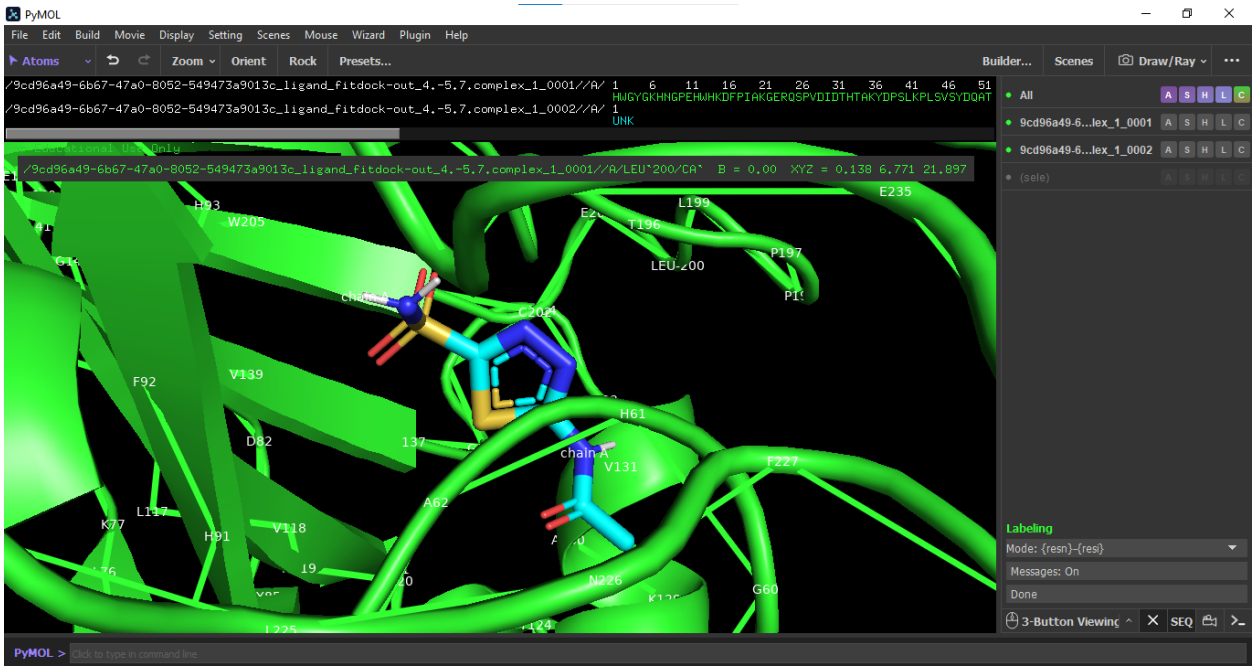
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File Edit Build Movie Display Setting Scenes Mouse Wizard Plugin Help
Atoms
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Orient
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Presets...
Builder...
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Draw/Ray
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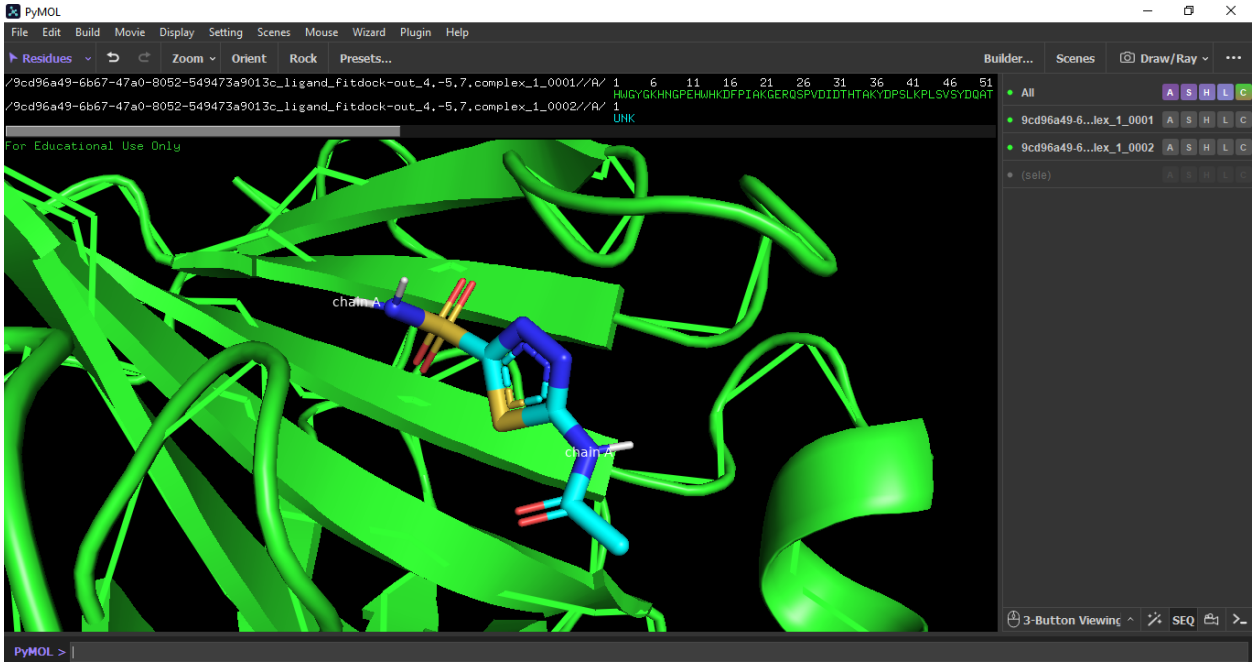
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Residues
Zoom
Orient
Rock
Presets...
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Draw/Ray
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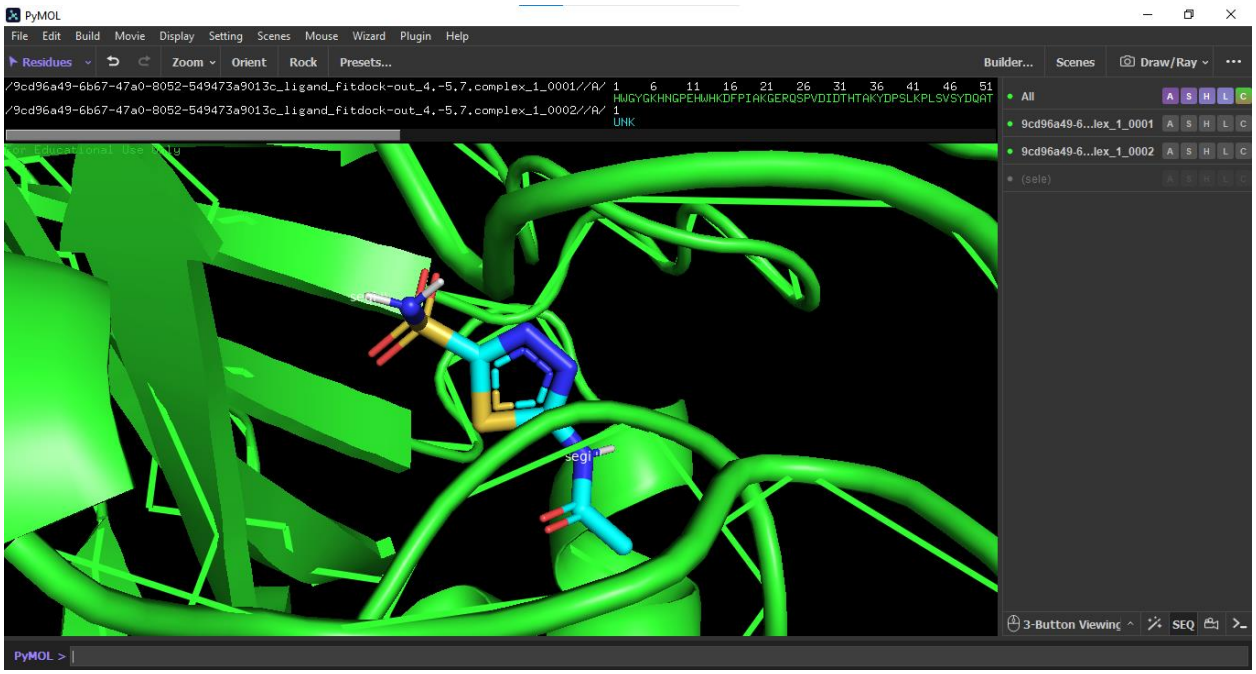
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Residues
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Rock
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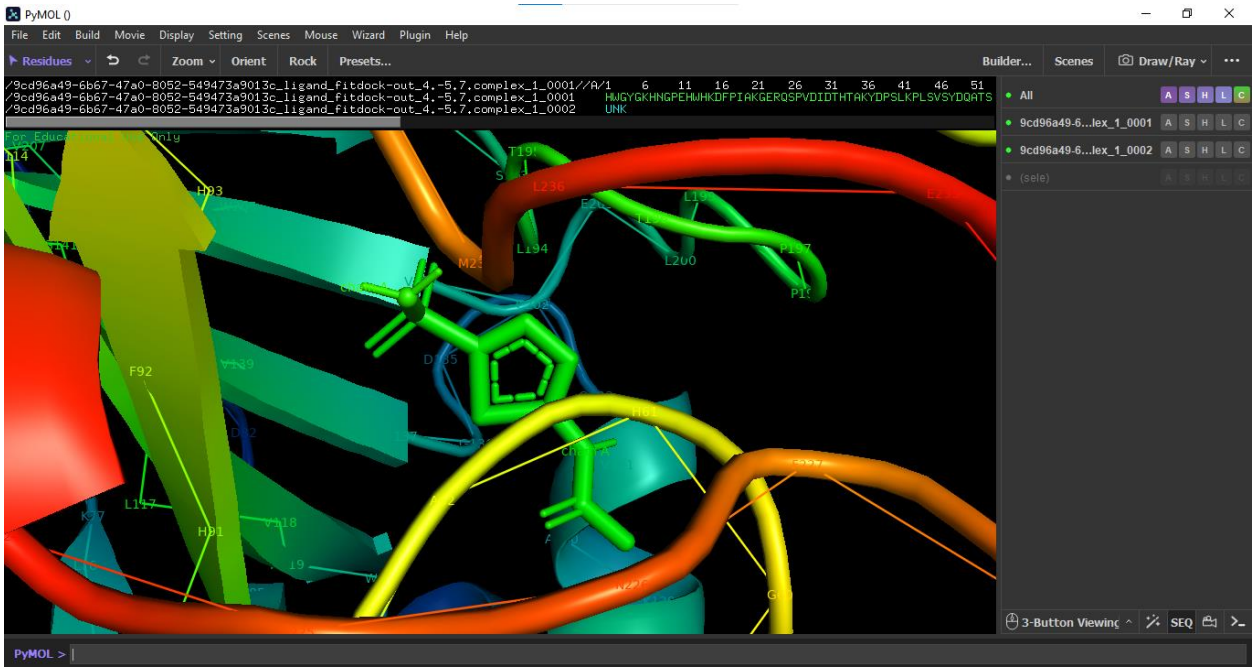
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Zoom
Orient
Rock
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